WRO®
WORLD ROBOT OLYMPIAD™

Team name	Compass
Team number	FE-002
Team members	Be Naro, Yeong Vechakasy Sothon, Art Oudom
Team coach	Taing Ly Min
Robotic set	Self-assembled set
Weight	1.10kg
Size	290× 175 × 180 mm
Building materials	Fully 3D-printed PLA chassis and mechanical parts
Controllers	Raspberry Pi 4, ESP32, L298N
Battery	12V, 3A battery pack; Raspberry Pi 4 uses separate USB power bank
Sensors	1× RPLiDAR A2M12, 1× 8 MP USB webcam
Motors	1× JGA25-370 DC gear motor, 1× MG996R servo
Pneumatic system	Not used
Spare parts	Battery x3, Battery charger x2, Screw driver set, Nuts x5, glue stick x1, power bank x1,
Programming Environment & language	Python, YOLO, ESP32 firmware
Drive & steering	RWD; single MG996R front steering servo; max steering angle 30°
Gear reduction	14 mm pinion → 25 mm gear; ≈1.79:1 reduction
Steering geometry	75 mm radius; 150 mm diameter; AB = 39.25 mm
Design iteration	3 chassis designs; LiDAR moved rear → front; electronics/battery moved to rear; camera mount raised after testing
LiDAR function	360° wall-distance and obstacle detection; primary lane/wall-following signal
Camera function	YOLO detects red/green pillars and provides secondary lane-position correction
Control algorithm	PID lane following + state machine combining LiDAR and vision
Red pillar	Keep pillar on the right
Green pillar	Keep pillar on the left
GitHub / Videos	https://youtu.be/hSMSpn8qdMk
Picture of the robot with start and stop bottom